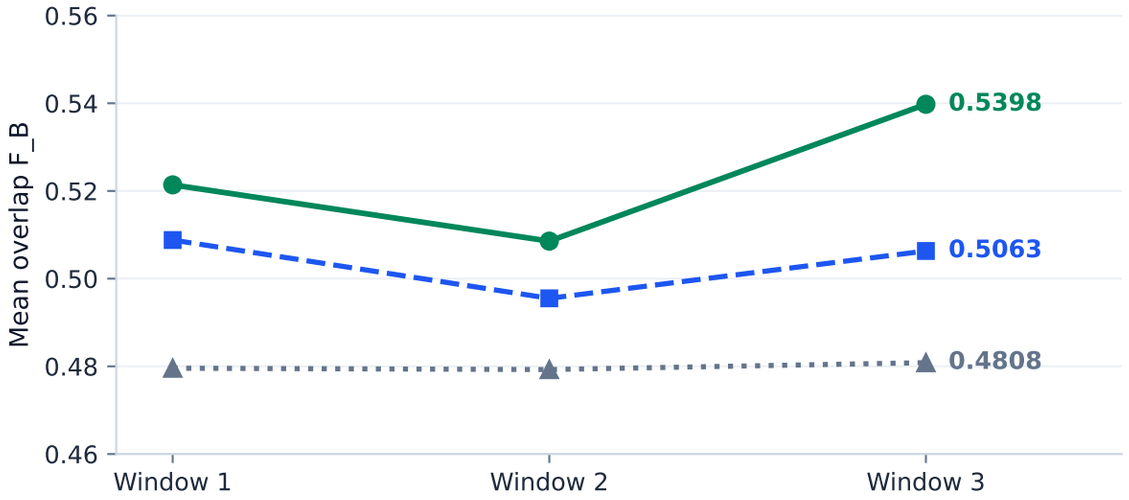
Prometheus: three-window hardware validation

Rigetti Cepheus-1-108Q | 18 logical workloads | 4-17 qubits | 221,184 shots

Metric throughout: classical distribution overlap F_B, higher is better.

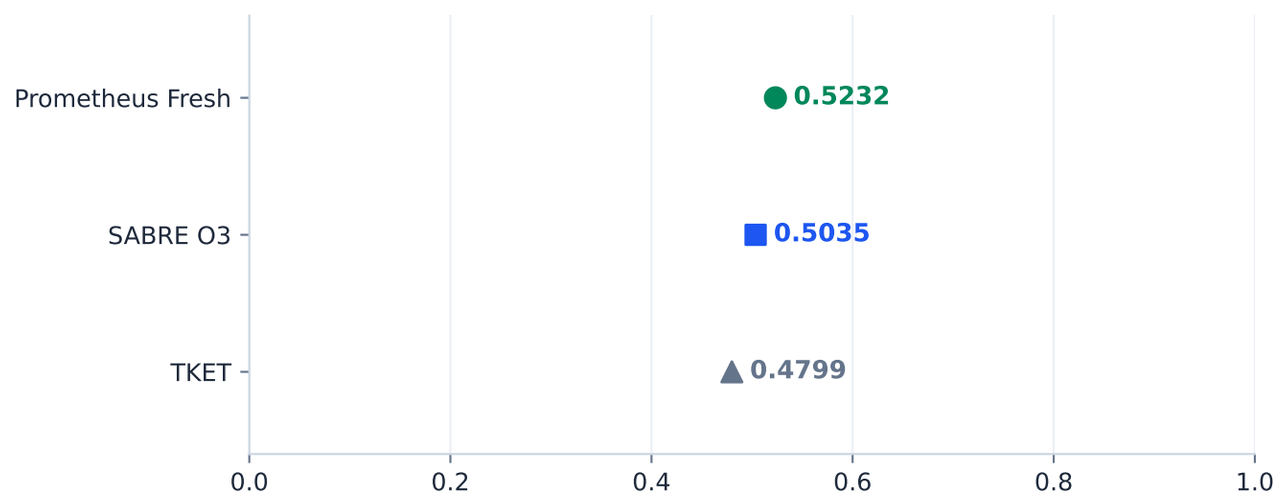
Prometheus Fresh SABRE O3 TKET

A Average performance in each window



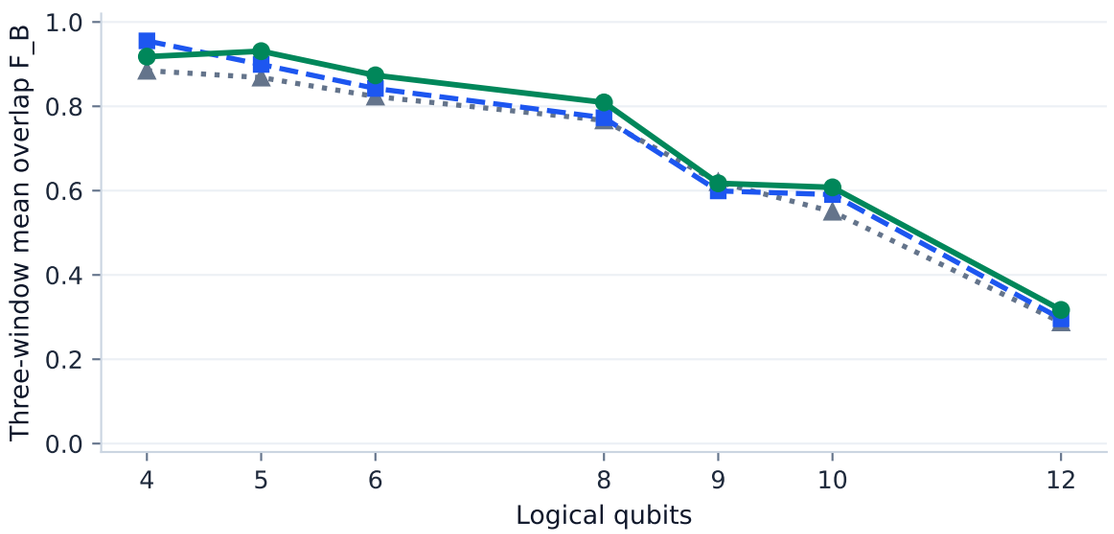
Focused vertical scale; points show observed means.

B Three-window pooled comparison



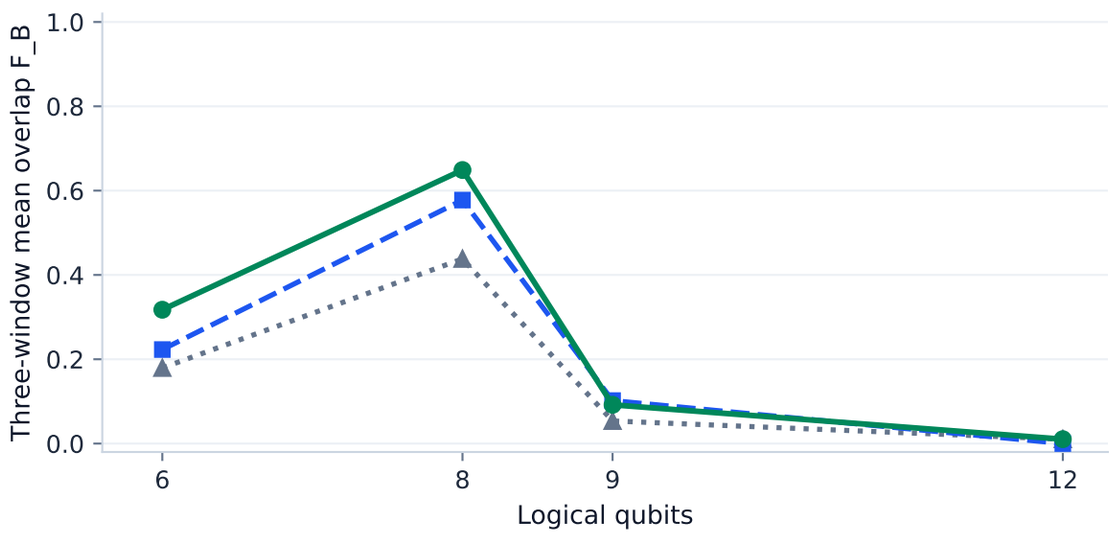
Each workload is averaged over its three executions.

C Dense all-to-all



One logical workload per plotted width; lines guide the eye.

D Phase-sensitive QFT



One logical workload per plotted width; lines guide the eye.

Pooled relative gain: +3.91% vs SABRE | +9.03% vs TKET

Exact paired sign-flip tests, Holm-adjusted across three comparisons: p = 0.0112 / 0.000641. N = 18 workloads (16W-0T-2L vs SABRE).

Control omitted for clarity. Panels A-B: all 18 workloads; C-D: named subsets. Pooled 2Q/depth reductions: -13.36%/-23.56% vs SABRE, -24.15%/-36.57% vs TKET.